Final CCP Project Report – AXON Fitness E- Commerce Website

1. Problem Complexity Analysis

Our e-commerce theme revolves around a gym- and fitness-focused website, introducing unique design and technical complexity.

The central challenge was the integration of a customized health-tracking system that evaluates a user's weight, lifestyle, and goals

to generate a personalized progress overview and diet/workout pathway. Unlike typical static product websites, our system required

dynamic inputs, conditional logic, and personalized guidance, demanding deeper design thinking and more advanced feature planning.

Additionally, ensuring seamless navigation, product listing consistency, and UI clarity within a fitness-oriented theme required careful structuring.

2. Unique Aspects Requiring Creative Problem- Solving

The most significant creative challenge was designing the custom progress-tracking and goal-setting feature.

We needed to conceptualize a system where users could input lifestyle details (active, sedentary, or highly active),

specify achievable goals, and receive feedback based on their capabilities rather than strict pre-defined plans.

This required creativity in planning both the user flow and technical logic so that the feature remained flexible,

user-friendly, and tailored to personal needs—unlike rigid templates found in existing fitness websites.

3. Multiple Solution Approaches Considered

Initially, we considered an e-commerce theme centered on groceries and daily essentials.

However, as we explored ideas related to healthy living, the concept evolved into a fitness-support website combining product sales

(dumbbells, whey protein, creatine) with personalized health insights.

We also evaluated different styling approaches. We chose pure CSS instead of Bootstrap because CSS allowed full creative freedom, custom design precision, and complete control over the layout. Bootstrap, although convenient, would have limited us with predefined components.

Our health-tracking feature was implemented through custom JavaScript to ensure adaptability and user customization.

4. Technical Challenges Encountered

One of the biggest difficulties was our lack of prior experience with HTML, CSS, and JavaScript.

We had to learn each technology from scratch, troubleshoot layout issues, and understand how files connect across the project structure.

Another challenge was writing separate files for HTML, CSS, and JS while ensuring precise linking and folder organization.

We overcame these issues through experimentation, online resources, and consistent practice.

Version control also introduced challenges. Setting up GitHub, linking VS Code, and resolving repository issues caused confusion initially.

With guidance from a senior, we learned branching, committing, and synchronizing code effectively.

5. Integration of Multiple Technologies

We followed a step-by-step process to ensure smooth integration. First, we generated AI images to maintain consistent branding.

Then we built the HTML structure, followed by CSS styling, and finally JavaScript functionality.

Git was configured before development to track progress and maintain backups.

To ensure component consistency, all project files were stored in a unified folder with structured sub-directories.

This ensured that images, pages, scripts, and stylesheets connected correctly across the website.

The integration of AI images, HTML/CSS, JavaScript, Git, and technical features resulted in a cohesive, intuitive user experience.

6. Performance & Trade- Off Analysis

Several trade-offs were made during development.

To maintain performance and avoid slow loading times, AI-generated images were optimized before use, even if this meant slightly reducing resolution.

We prioritized clean CSS over framework-heavy designs to ensure faster performance and smaller file sizes.

The custom technical feature added complexity to the JavaScript logic, but it offered significantly better personalization.

Overall, we balanced design quality, performance, and development efficiency to deliver a polished and functional final product.